```
from google.colab import files

upload = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

PROJECT TITLE

SIMPLE LINEAR REGRESSION - MARKETING ROI ANALYSIS.

PROJECT OVERVIEW This project analyzes the relationship between marketing expenditure and Sales using Simple Linear Regression. The objective is to identify the marketing channel that has the strongest impact on sales, build an OLS regression model using Statsmodels, validate the model assumptions, and provide business recommendations based on the findings.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
```

```
import pandas as pd

df = pd.read_csv("85334965-5736-457a-b8d4-a077e6872f84.csv")
df.head()
```

	TV	Radio	Social_Media	Sales
0	16.0	6.566231	2.907983	54.732757
1	13.0	9.237765	2.409567	46.677897
2	41.0	15.886446	2.913410	150.177829
3	83.0	30.020028	6.922304	298.246340
4	15.0	8.437408	1.405998	56.594181

DATA EXPLORATION & CLEANING

In this section, the dataset is explored to understand its structure, check for missing values, and prepare it for analysis.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 4546 entries, 0 to 4571
Data columns (total 4 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   TV            4546 non-null   float64
 1   Radio         4546 non-null   float64
 2   Social_Media  4546 non-null   float64
 3   Sales         4546 non-null   float64
dtypes: float64(4)
memory usage: 177.6 KB
```

```
df.describe()
```

	TV	Radio	Social_Media	Sales
count	4546.000000	4546.000000	4546.000000	4546.000000
mean	54.062912	18.157533	3.323473	192.413332
std	26.104942	9.663260	2.211254	93.019873
min	10.000000	0.000684	0.000031	31.199409
25%	32.000000	10.555355	1.530822	112.434612
50%	53.000000	17.859513	3.055565	188.963678
75%	77.000000	25.640603	4.804919	272.324236
max	100.000000	48.871161	13.981662	364.079751

```
df.isnull().sum()
```

	0
TV	0
Radio	0
Social_Media	0
Sales	0

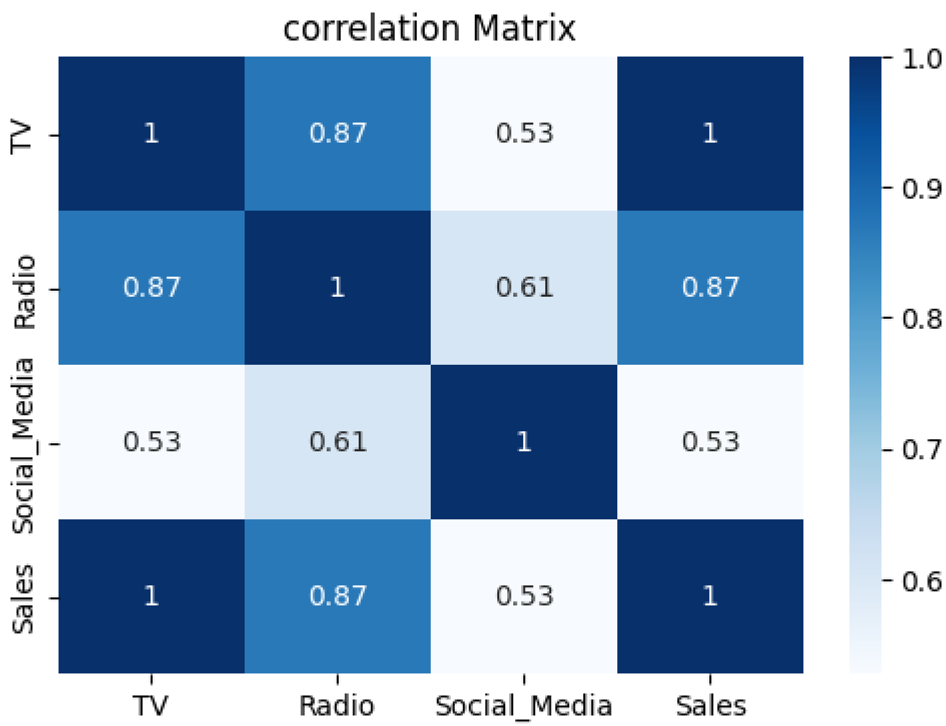
dtype: int64

```
df = df.dropna()
```

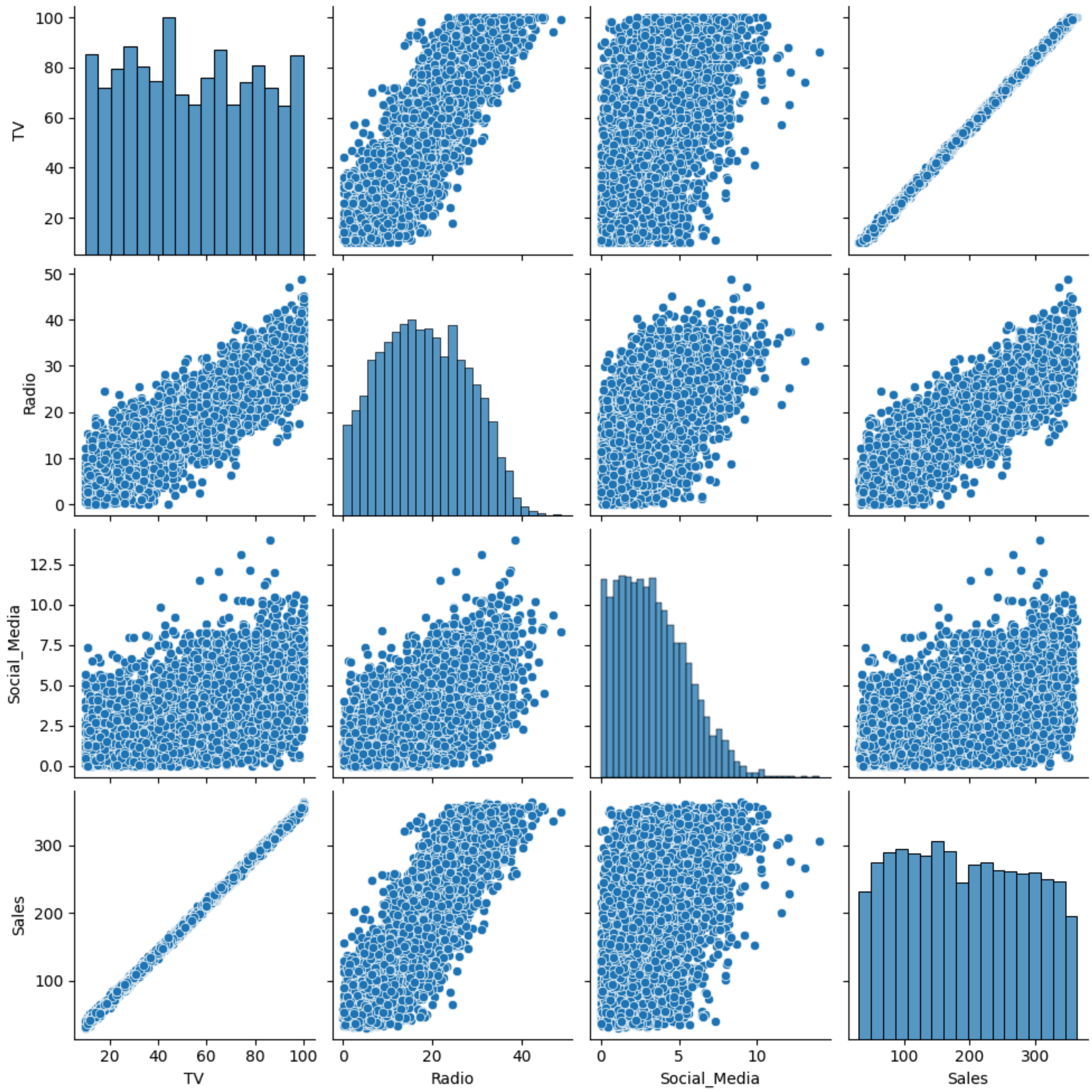
```
df.corr(numeric_only=True)
```

	TV	Radio	Social_Media	Sales
TV	1.000000	0.869158	0.527687	0.999497
Radio	0.869158	1.000000	0.606338	0.868638
Social_Media	0.527687	0.606338	1.000000	0.527446
Sales	0.999497	0.868638	0.527446	1.000000

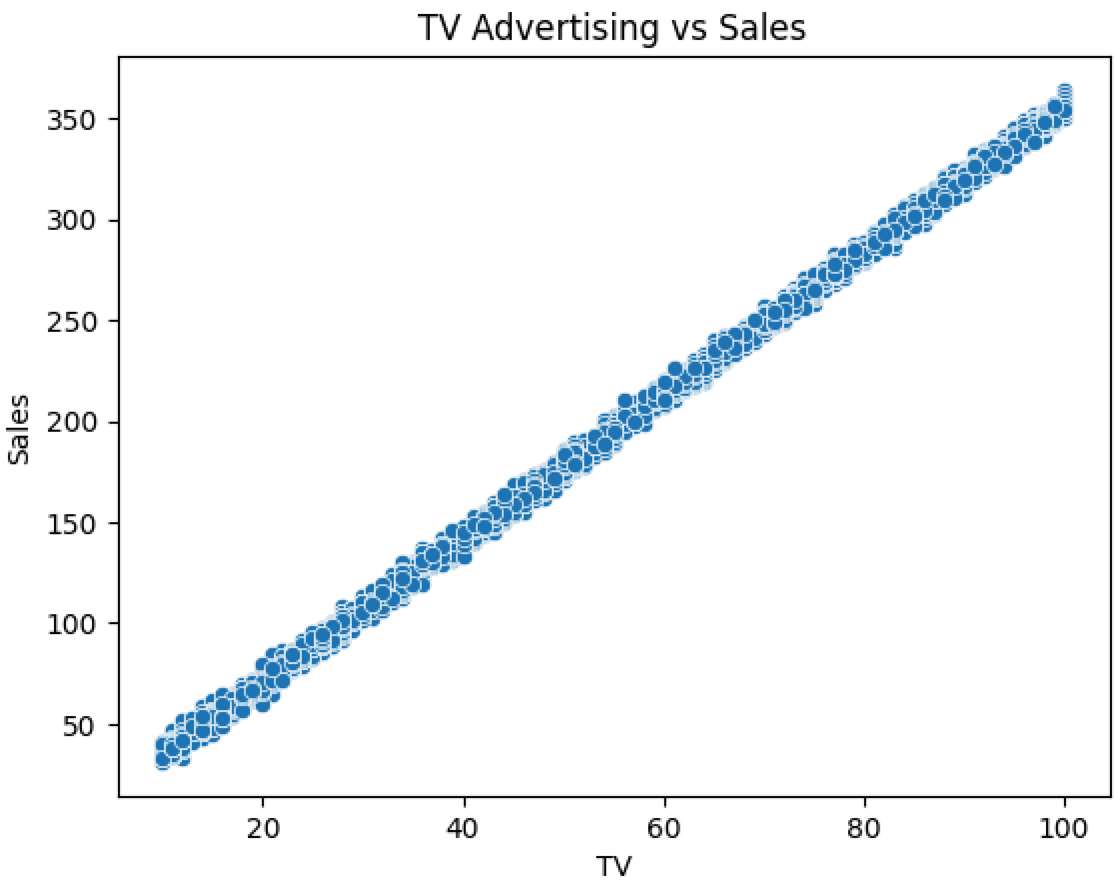
```
plt.figure(figsize=(6,4))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="Blues")
plt.title("correlation Matrix")
plt.show()
```




```
sns.pairplot(df)
plt.show()
```


TV
Radio
Social_Media
Sales
100
80
60
40
20
50
40
30
20
10
0
12.5
10.0
7.5
5.0
2.5
0.0
300
200
100
20
40
60
80
100
0
20
40
0
5
10
100
200
300
TV
Radio
Social_Media
Sales

```
sns.scatterplot(x="TV", y="Sales", data=df)
plt.title("TV Advertising vs Sales")
plt.show()
```



VARIABLE SELECTION

The correlation analysis shows that TV advertising has the strongest relationship with Sales.

Correlation values:

- TV = 0.9995
- Radio = 0.8691
- Social_Media = 0.5289

Therefore, TV is selected as the independent variable for the Simple Linear Regression model.

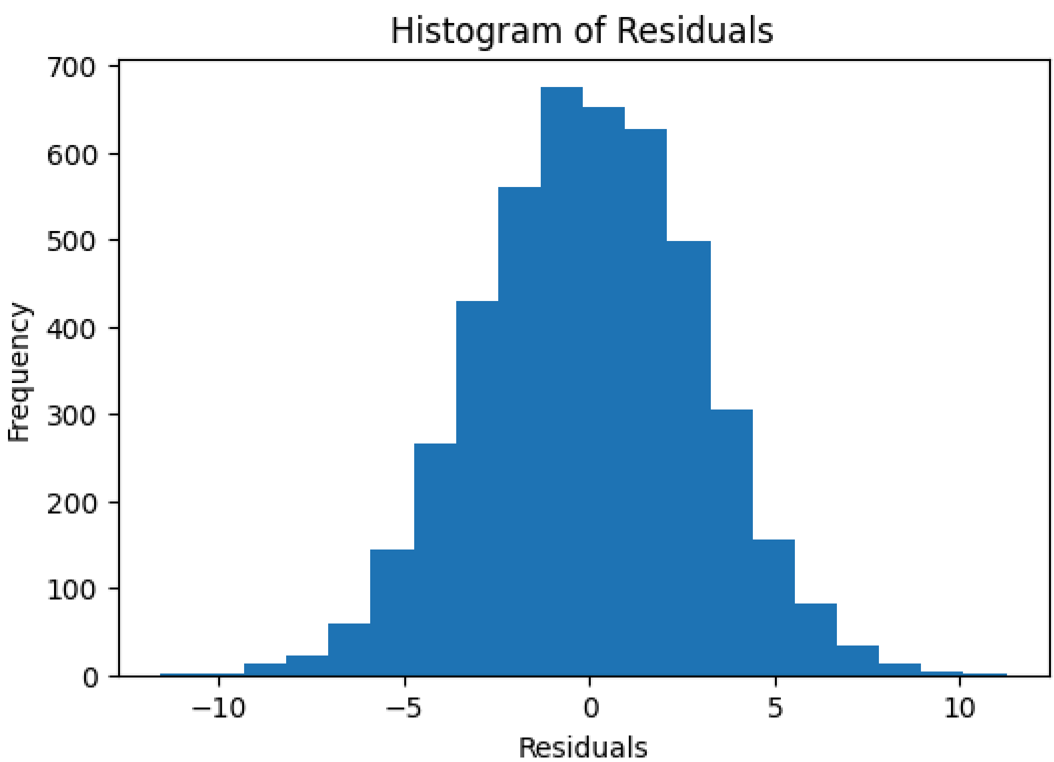
```
Y = df['Sales']

x = sm.add_constant(x)

model = sm.OLS(Y, x).fit()

print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
```

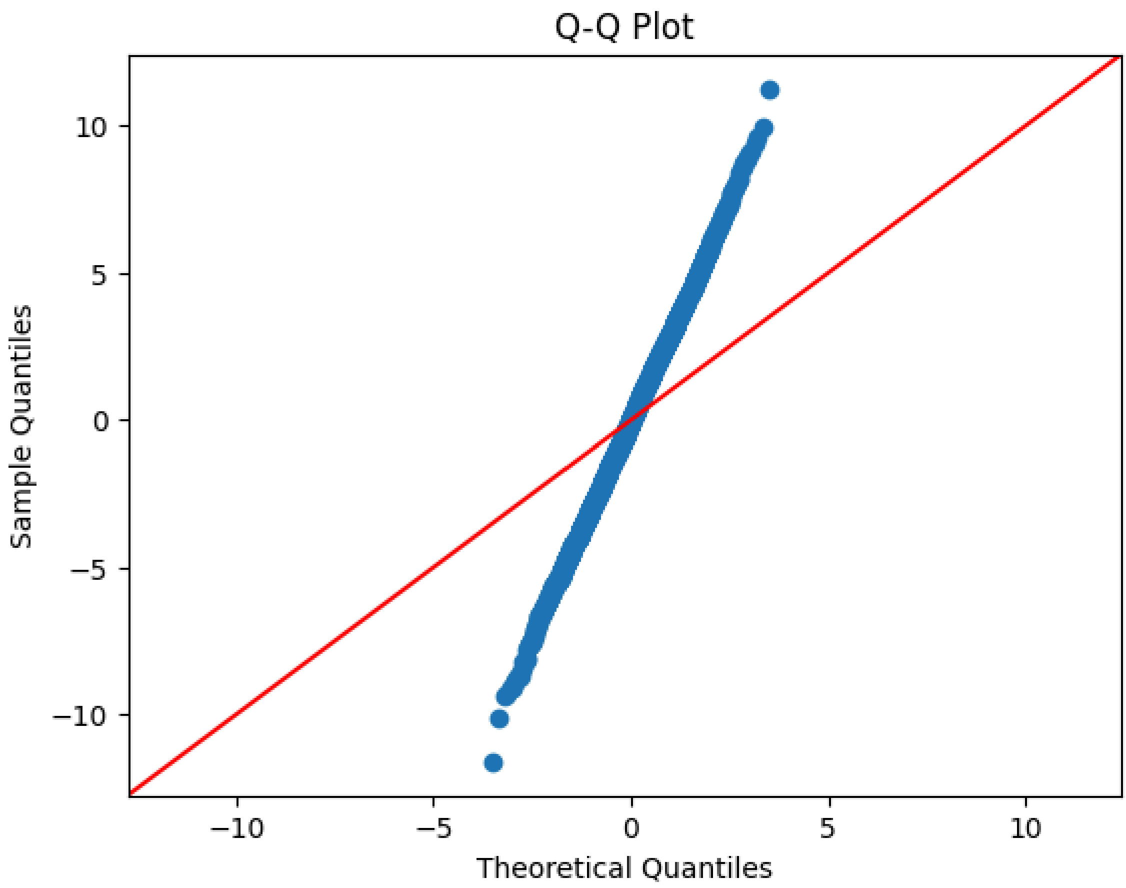
```
Dep. Variable:                  Sales   R-squared:                       0.99
Model:                            OLS   Adj. R-squared:                  0.99
Method:                 Least Squares   F-statistic:                 4.517e+0
Date:                Fri, 26 Jun 2026   Prob (F-statistic):               0.0
Time:                        11:32:56   Log-Likelihood:                -11366
No. Observations:                4546   AIC:                         2.274e+0
Df Residuals:                    4544   BIC:                         2.275e+0
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975
------------------------------------------------------------------------------
const         -0.1325      0.101     -1.317      0.188      -0.330       0.06
TV             3.5615      0.002   2125.272      0.000       3.558       3.56
==============================================================================
Omnibus:                        0.052   Durbin-Watson:                   1.99
Prob(Omnibus):                  0.974   Jarque-Bera (JB):                0.03
Skew:                          -0.001   Prob(JB):                        0.98
Kurtosis:                       3.012   Cond. No.                         138
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correc
```

```
residuals = model.resid
fitted = model.fittedvalues
```

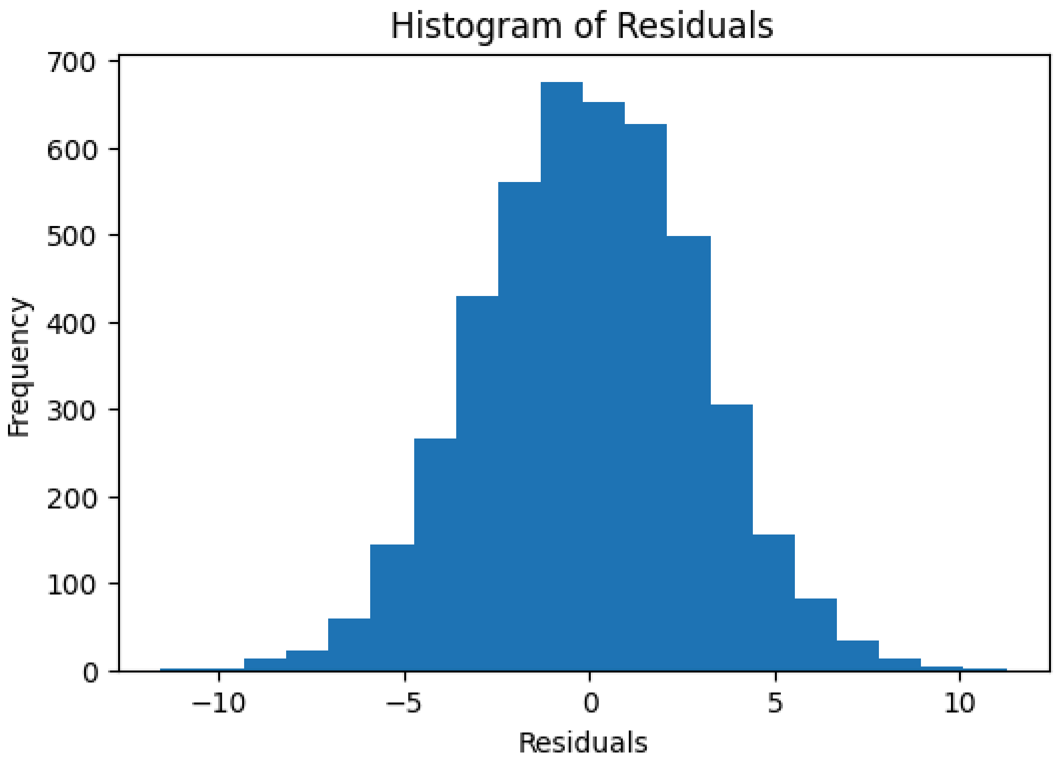
```
plt.figure(figsize=(6,4))
plt.hist(residuals, bins=20)
plt.title("Histogram of Residuals")
plt.xlabel("Residuals")
plt.ylabel("Frequency")
plt.show()
```



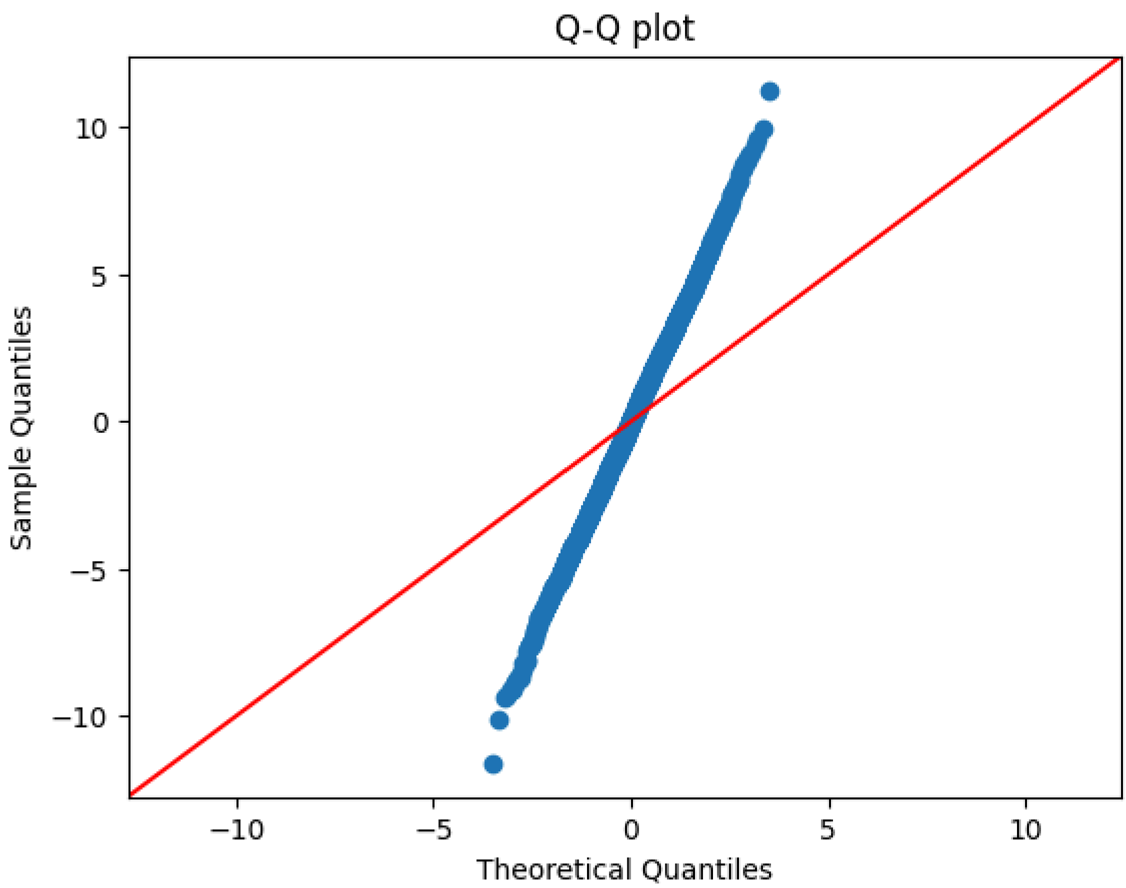
```
sm.qqplot(residuals, line='45')
plt.title("Q-Q Plot")
plt.show()
```




```
plt.figure(figsize=(6,4))
plt.hist(residuals, bins=20)
plt.title("Histogram of Residuals")
plt.xlabel("Residuals")
plt.ylabel("Frequency")
plt.show()
```

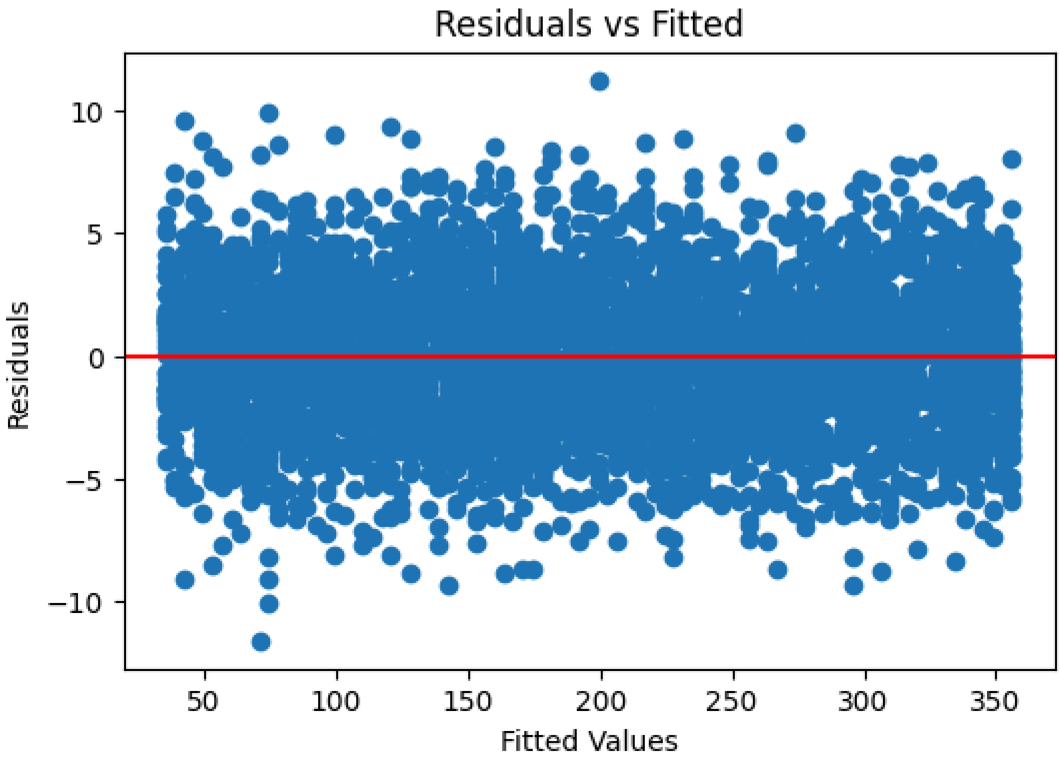



```
sm.qqplot(residuals, line='45')
plt.title("Q-Q plot")
plt.show()
```




```
plt.figure(figsize=(6,4))
plt.scatter(fitted, residuals)
```

```
plt.axhline(y=0, color='red')
plt.xlabel("Fitted Values")
plt.ylabel("Residuals")
plt.title("Residuals vs Fitted")
plt.show()
```



INTERPRETATION & EVALUATION

R-squared The model achieved an R-squared value of 0.999. This means that approximately 99.9% of the variation in Sales is explained by TV advertising expenditure, indicating an extremely strong relationship between the variables.

COEFFICIENT INTERPRETATION The coefficient for TV advertising is 3.5615. This means that for every one - unit increase in TV advertising expenditure, Sales are expected to increase by approximately 3.56 units, holding all other factors constant.

P - VALUE INTERPRETATION. The p-value for TV is 0.000, which is less than the significance level of 0.05. This indicates that the relationship between TV advertising and Sales is statistically significant and unlikely to have occurred by chance.

INTERCEPT INTERPRETATION. The intercept is -0.1325. This represents the predicted Sales value when TV advertising expenditure is zero.

MODEL EVALUATION. The high R-squared value and statistically significant p-value suggest that TV advertising is a strong predictor of Sales and that the regression